Local zero-free shapes from reflected logarithmic-derivative kernels

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Abstract

We construct local additions to a zero-free half-plane for the Riemann zeta function, conditional on the global bound $\beta \le 7/8$ for its nontrivial zeros. Positivity of the Hadamard real-part expansion to the right of this bound gives a single-zero disk. Keeping the same-height reflected zero gives a larger exclusion region described by a quartic inequality. An external Arb computation at $219/250 + 4 \cdot 10^{12} i$ supplies the upper bound $29/2$ for the real logarithmic derivative of the completed function. Exact rational corner values and concavity then exclude every point with $\operatorname{Re} s \ge 4/5$ and $|\operatorname{Im} s - 4 \cdot 10^{12}| \le 1/200$. The real algebra and conditional exclusion implications are formalized in Lean. The global zero bound, the analytic kernel identity, and the Arb evaluation remain separate inputs. We also give a finite nonnegative multi-anchor certificate, with its pointwise limitation made explicit. The result is a local shape extension; no new uniform strip or end-to-end formalized zeta theorem is claimed.

1 Statement and scope

Let

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s). \tag{1}$$

This is the entire completed Riemann function. Its zeros are precisely the nontrivial zeros of ζ, with the same multiplicities. Throughout the paper, a sum over zeros includes all ordinates of both signs and includes multiplicities.

Our global analytic hypothesis is

$$\zeta(\beta + i\gamma) = 0, \quad 0 < \beta < 1 \quad \Longrightarrow \quad \beta \le \frac{7}{8}. \tag{$H_{7/8}$}$$

The seven-eighths input and the stronger cubic boundary studied in [6] are used only with their stated proof dependencies. In particular, this paper does not supply the imported arithmetic probe obligation of that work. The local construction below needs only ($H_{7/8}$); it is independent of the small improvement from 7/8 to the cubic boundary.

Fix the exact constants

$$\sigma_c = \frac{219}{250}, \qquad T_0 = 4000000000000, \qquad E = \frac{29}{2}, \qquad K = \operatorname{Re} \frac{\xi'}{\xi}(\sigma_c + iT_0). \tag{3}$$

Notice that $\sigma_c - 7/8 = 1/1000 > 0$. Thus ($H_{7/8}$) ensures that $\xi(\sigma_c + iT_0) \ne 0$ and that K is well defined.

Theorem 1.1 (Conditional local rectangle)**.** *Assume ($H_{7/8}$) and $K \le E$, with the constants in (3). Then*

$$\zeta(s) \ne 0 \qquad \text{if} \qquad \operatorname{Re} s \ge \frac{4}{5}, \qquad |\operatorname{Im} s - 4000000000000| \le \frac{1}{200}. \tag{4}$$

The same conclusion holds with the central ordinate replaced by -4000000000000. All displayed rectangle edges are included.

Section 5 records a rigorous external ball-arithmetic calculation of $K < E$, conditional only on accepting the correctness of the cited numerical library and the executable evaluation. Thus the numerical instance of Theorem 1.1 still depends on ($H_{7/8}$). Theorem 3.2 gives the curved region of which (4) is a particularly simple rational subset.

The choice of T_0 is an illustration at a high ordinate. It exceeds the height 3000175332800 covered by the computation in Platt–Trudgian [5]; we make no assertion about the largest verified height at the date of this paper. Our calculation does not enumerate zeros or verify the Riemann hypothesis up to T_0.

2 Positive kernels from the completed function

We first recall the analytic identity underlying the construction. Related real-part positivity questions are discussed in Goldštein–Grigutis [1]. The identity used here comes from the classical Hadamard product; the local geometric consequences will be derived explicitly.

Lemma 2.1 (Real-part Hadamard identity)**.** *For any $s = \sigma + it$ with $\xi(s) \neq 0$,*

$$\operatorname{Re} \frac{\xi'}{\xi}(s) = \sum_{\rho=\beta+i\gamma} \frac{\sigma - \beta}{(\sigma-\beta)^2 + (t-\gamma)^2}. \tag{5}$$

The real series on the right is absolutely convergent.

Proof. The order-one Hadamard product gives

$$\frac{\xi'}{\xi}(s) = b_0 + \sum_{\rho} \left(\frac{1}{s-\rho} + \frac{1}{\rho} \right)$$

with the canonical summation convention. For fixed s and large $|\gamma|$, the real part of each of $1/(s-\rho)$ and $1/\rho$ is $O_s(\gamma^{-2})$, since $0 < \beta < 1$. The classical zero count $N(T) = O(T \log T)$ therefore gives absolute convergence of the separate real-part series. Taking real parts leaves a possible real constant

$$C = \operatorname{Re} b_0 + \sum_{\rho} \operatorname{Re} \frac{1}{\rho}.$$

This constant is zero. Indeed, the functional equation $\xi(s) = \xi(1-s)$ gives $\xi'(1/2) = 0$, while $\xi(1/2) \neq 0$. The latter is also immediate from the alternating-series expression for $\zeta(1/2)$, which is negative. At $s = 1/2$, the terms $\operatorname{Re}(1/(1/2-\rho))$ cancel under $\rho \mapsto 1-\overline{\rho}$; critical-line terms are zero. Absolute convergence justifies this pairing. Hence $C = 0$, proving (5). This argument also shows that no additional real constant or factor of two belongs in (5). □

More generally, assume that every nontrivial zero satisfies $\beta \leq B$, where $1/2 \leq B < 1$, and choose an anchor $s_c = \sigma_c + iT$ with $\sigma_c > B$. Every summand in (5) is then strictly positive. For an individual zero,

$$\frac{\sigma_c - \beta}{(\sigma_c-\beta)^2 + (T-\gamma)^2} \leq \operatorname{Re} \frac{\xi'}{\xi}(s_c). \tag{6}$$

This use of positivity needs no truncation or estimate of the remaining zero sum.

Proposition 2.2 (Single-zero disk)**.** *Under the preceding global bound, suppose* $\operatorname{Re}(\xi'/\xi)(\sigma_c + iT) \leq E$ *for a real* $E > 0$*. Then every point of the open disk*

$$\left(\sigma - \left(\sigma_c - \frac{1}{2E} \right) \right)^2 + (t-T)^2 < \frac{1}{4E^2} \tag{7}$$

is not a nontrivial zero of ζ.

Proof. Set $a = \sigma_c - \sigma$ and $u = t - T$. The strict disk inequality is $a^2 + u^2 < a/E$. It implies $a > 0$ and $a^2 + u^2 > 0$; in particular the anchor, where $a = u = 0$, is not inside the disk. Dividing gives $a/(a^2 + u^2) > E$, contrary to (6) for a zero. □

For (3), this disk has radius $1/29$, real center $6101/7250$, and leftmost real coordinate $5851/7250$ ($0.8070344827\ldots$). The reflected-pair construction below reaches further left.

3 The reflected-pair quartic region

Lemma 3.1 (Two distinct same-height zeros)**.** *Let $\rho = \beta + i\gamma$ be a nontrivial zero with $\beta > 1/2$. Then $1 - \overline{\rho} = 1 - \beta + i\gamma$ is a distinct zero of the same multiplicity. Under the global bound $\beta \le B < \sigma_c$,*

$$\Phi_{\sigma_c,T}(\beta,\gamma) := \frac{\sigma_c - \beta}{(\sigma_c - \beta)^2 + (\gamma - T)^2} + \frac{\sigma_c + \beta - 1}{(\sigma_c + \beta - 1)^2 + (\gamma - T)^2} \le \operatorname{Re} \frac{\xi'}{\xi}(\sigma_c + iT). \tag{8}$$

Proof. Complex conjugation and the functional equation of ξ preserve zero multiplicities. Their composition gives the stated same-height reflection. Distinctness follows from $\beta > 1/2$. Both terms in (8) occur in the positive sum (5). Even when a zero has multiplicity greater than one, taking one copy of each gives the stated inequality; taking all copies could only strengthen it. Critical-line zeros would coincide with their reflected zeros and cannot be counted twice by this argument. □

For $1/2 < \sigma \le B$ put

$$a = \sigma_c - \sigma > 0, \quad b = \sigma_c + \sigma - 1 > 0, \quad c = a + b = 2\sigma_c - 1, \quad p = ab > 0, \quad z = (t-T)^2 \ge 0. \tag{9}$$

The sum of the two kernels simplifies to

$$\frac{a}{a^2 + z} + \frac{b}{b^2 + z} = \frac{c(p + z)}{(p - z)^2 + c^2 z}. \tag{10}$$

In fact, the common denominator is

$$(a^2 + z)(b^2 + z) = (p - z)^2 + c^2 z > 0,$$

and the numerator is $(a + b)(ab + z)$. The strict positivity of the denominator also follows directly from $a, b > 0$ and $z \ge 0$.

Theorem 3.2 (Quartic shape certificate)**.** *Assume that all nontrivial zeros satisfy $\beta \le B$, where $1/2 \le B < 1$ and $\sigma_c > B$, and that $\operatorname{Re}(\xi'/\xi)(\sigma_c + iT) \le E$ for $E > 0$. Define*

$$S_E(p, z) = c(p + z) - E\big((p - z)^2 + c^2 z\big), \qquad c = 2\sigma_c - 1. \tag{11}$$

Then the region

$$\frac{1}{2} < \sigma \le B, \qquad S_E\big((\sigma_c - \sigma)(\sigma_c + \sigma - 1), (t - T)^2\big) > 0 \tag{12}$$

contains no zeros of ζ. Its union with $\{\operatorname{Re} s > B\}$ contains no nontrivial zeros. The latter half-plane contains the pole $s = 1$, so it is a zero-free meromorphic domain rather than a domain on which ζ is everywhere finite.

Proof. If a zero lay in (12), then (10), the positive denominator, and $S_E > 0$ would imply $\Phi_{\sigma_c,T}(\sigma, t) > E$. This contradicts Lemma 3.1 and the assumed upper bound. The half-plane assertion follows from the global bound. □

The polynomial in (12) has total degree four in σ and $t-T$. This degree arises from two positive kernel terms; it is not a fourth-moment estimate for ζ or for a detector trace.

One can describe its horizontal slices without approximating a boundary curve. For fixed $z \geq 0$,

$$S_E(p,z) = -Ep^2 + (c+2Ez)p + cz - Ez^2 - Ec^2z.$$

The discriminant is

$$D(z) = c^2 + 4Ecz(2-Ec). \tag{13}$$

If $D(z) > 0$, the condition is precisely

$$p_-(z) < p < p_+(z), \qquad p_\pm(z) = \frac{c+2Ez \pm \sqrt{D(z)}}{2E}. \tag{14}$$

If $D(z) \leq 0$, no strict positive slice exists. These statements are only to be used together with $1/2 < \sigma \leq B$ and $p > 0$. At $z = 0$, (14) reduces to $0 < p < c/E$. For the constants (3), the corresponding left edge at the central ordinate is

$$\frac{1}{2} + \sqrt{\left(\frac{219}{250} - \frac{1}{2}\right)^2 - \frac{94/125}{29/2}} = 0.7991887882\ldots. \tag{15}$$

The exact radical, rather than this illustrative decimal, specifies the edge. Strict inequality excludes points to its right within the quartic domain.

Remark 3.3 (Do not fill unproved gaps)**.** A quartic component can be separated from the old half-plane at some heights. The certified region is the union actually stated in Theorem 3.2; it is not automatically the entire half-plane to the right of the leftmost point of each slice. For the smaller ordinate window in Theorem 1.1, Section 4 proves positivity all the way from 4/5 to 7/8, so there is no such gap across that rectangle.

4 An exact rational rectangle certificate

We now specialize to (3), so $c = 94/125$, and write $S = S_E$. For $4/5 \leq \beta \leq 7/8$ define

$$p(\beta) = \left(\frac{219}{250} - \beta\right)\left(\frac{219}{250} + \beta - 1\right), \qquad z = (\gamma - T_0)^2.$$

The exact intervals needed are

$$p_-^{\mathrm{box}} = \frac{751}{1000000} \leq p \leq p_+^{\mathrm{box}} = \frac{3211}{62500}, \qquad 0 \leq z \leq z_+^{\mathrm{box}} = \frac{1}{40000}. \tag{16}$$

These bounds require no numerical monotonicity check. Indeed,

$$p(\beta) - \frac{751}{1000000} = \left(\frac{7}{8} - \beta\right)\left(\beta - \frac{1}{8}\right) \geq 0,$$
$$\frac{3211}{62500} - p(\beta) = \left(\beta - \frac{4}{5}\right)\left(\beta - \frac{1}{5}\right) \geq 0.$$

The height bound $|\gamma - T_0| \leq 1/200$ gives the bound on z.

Lemma 4.1 (Positivity on the entire parameter box)**.** *For every real (p,z) in (16), $S(p,z) > 0$.*

Proof. The four values are the following positive rational numbers:

p	z	$S(p,z)$
751/1000000	0	1113147971/2000000000000
751/1000000	1/40000	185457099/500000000000
3211/62500	0	2828891/7812500000
3211/62500	1/40000	426282771/2000000000000

The quadratic part of S is $-E(p-z)^2$. Its Hessian is

$$-2E\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix},$$

which is negative semidefinite. Thus S is jointly concave. Every point of the box is a convex combination of its four corners, so concavity bounds S below by the corresponding convex combination of the corner values. In particular,

$$S(p,z) \geq \frac{426282771}{2000000000000} > 0. \tag{17}$$

The implemented Lean proof reaches strict positivity by concave quadratic interpolation first in p and then in z; it proves positivity for all real points, not only for rational samples. □

Proof of Theorem 1.1. Suppose that a zero $s = \beta + i\gamma$ satisfied the stated rectangle conditions. Since $\beta \geq 4/5 > 0$, it cannot be a trivial zero. Nontrivial zeros have $\beta < 1$ and, by ($H_{7/8}$), $\beta \leq 7/8$. Equations (16) and Lemma 4.1 therefore place it in the region of Theorem 3.2, which is impossible. The proof uses strict positivity even at all four parameter-box corners, hence the closed edges of the original rectangle are covered. The identity $\zeta(\overline{s}) = \overline{\zeta(s)}$ gives the negative-height copy. □

5 External numerical evaluation and reproducibility

Directly evaluating ξ itself at the chosen ordinate is unnecessary. Differentiating (1) gives the exact identity

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi(s/2) + \frac{\zeta'(s)}{\zeta(s)}, \tag{18}$$

where $\psi = \Gamma'/\Gamma$. All displayed terms are retained in the executable evaluation.

Proposition 5.1 (External Arb enclosure)**.** *The supplied evaluation with Python 3.12.5, python-flint 0.9.0, and linked FLINT 3.6.0 returns a real ball for the value K in (3) contained in the rational open interval*

$$\frac{14495344672513496 99}{10^{17}} < K < \frac{1449534467251349700}{10^{17}} < \frac{29}{2}. \tag{19}$$

This is a rigorous external arithmetic certificate, not a Lean kernel theorem.

Computational verification. Arb uses real and complex ball arithmetic with rigorous error tracking [2, 3]. The anchor is supplied as the exact integer ordinate and a ball enclosing the exact rational real part. A degree-one Taylor jet of $\zeta(s+X)$, computed with rigorously bounded derivatives [4], produces both $\zeta(s)$ and $\zeta'(s)$ in one calculation. The program checks that the enclosure for $\zeta(s)$ does not contain zero before dividing. Formula (18) then gives a ball for K.

The evaluation is recomputed at 96 and 160 bits. Each run checks strict interval separation from both rational endpoints in (19) and from 29/2. At 160 bits the recorded real ball is

$$\left[14.4953446725134969914321472931200539554146 37 \;\pm\; 9.02\cdot 10^{-43}\right].$$

The full complex enclosures, precision settings, library version, and successful rational comparisons are stored in the supplied JSON record. The two precisions are replays of the same library; they are not independent implementations of the special functions. □

The repository artifacts use the following paths:

Artifact	Scope
`scripts/local_shape_certificate.py`	Arb evaluation and exact rational geometry checks.
`verification/local-shape-arb.json`	Recorded balls, exact constants, and corner values.
`ZeroFree/LocalShape.lean`	Kernel-checked real algebra and conditional exclusions.

With python-flint installed, reproduce the numerical calculation from the repository root by running

```
python scripts/local_shape_certificate.py
```

The script's optional plot is illustrative; the certified geometry is the exact inequality and rational rectangle, rather than a sampled contour or a decimal boundary.

6 The formalization boundary

The Lean module works with real variables and a generic predicate $\mathsf{IsZero}(\beta, t)$. It does not define ζ or ξ. Its analytic interface is an explicit hypothesis saying that a putative zero obeys the paired kernel majorization with upper bound E. In the mathematical application, Lemma 3.1 and $K \le E$ supply this interface.

The module proves the exact four corner identities, positivity throughout the real parameter box, the bounds taking a physical rectangle into that box, the positive denominator, and the identity (10). It then proves the generic quartic exclusion, the rectangle exclusion, and the rectangle exclusion with an explicit global-bound hypothesis. The single-kernel open disk is also checked. The module additionally proves finite nonnegative weighted budget inequalities, generic conditional weighted exclusions, their reflected-pair specialization, and the pointwise union limitation of Proposition 7.2. Each analytic hypothesis is a theorem parameter; no analytic claim is inserted as an axiom declaration.

The axiom reports list only `propext`, `Classical.choice`, and `Quot.sound`. The module has no `sorry`, `sorryAx`, or `native_decide`. These checks establish the finite algebra and the conditional implications. They do not certify the correctness of the external Arb library, formalize the Hadamard product, or discharge ($H_{7/8}$). A future end-to-end Lean theorem for the concrete zeta function must connect and prove those separate inputs.

The written joint-concavity margin (17), the general symbolic parameters of Theorem 3.2, and the radical slice description are written analytic or algebraic results of this paper. The current Lean module certifies the fixed constants, strict real-box positivity, and abstract finite weighted implications described above. In particular, the analytic bridge from the concrete zeta function to the multi-anchor kernel budgets remains external. The module does not claim to formalize every generalized statement in the paper.

7 Finite multi-anchor certificates and limitations

The basic argument can use any finite collection of certified anchors. This is useful when a common rational inequality is easier to check on a whole box than a case split between individual anchors.

Theorem 7.1 (Nonnegative multi-anchor certificate)**.** *Assume that all nontrivial zeros satisfy $\beta \le B$, where $1/2 \le B < 1$. For $1 \le j \le m$ choose $s_j = \sigma_j + iT_j$ with $\sigma_j > B$, and bounds*

$$K_j = \operatorname{Re} \frac{\xi'}{\xi}(s_j) \le E_j.$$

Let $w_j \ge 0$ be real weights, and set

$$\Phi_j(\sigma, t) = \frac{\sigma_j - \sigma}{(\sigma_j - \sigma)^2 + (t - T_j)^2} + \frac{\sigma_j + \sigma - 1}{(\sigma_j + \sigma - 1)^2 + (t - T_j)^2}.$$

Then the region

$$\frac{1}{2} < \sigma \le B, \qquad \sum_{j=1}^{m} w_j \Phi_j(\sigma, t) > \sum_{j=1}^{m} w_j E_j \tag{20}$$

contains no zeros of ζ.

Proof. For a hypothetical zero in the stated real-part range, Lemma 3.1 at every anchor gives $\Phi_j(\sigma, t) \le K_j \le E_j$. Multiplication by nonnegative weights and finite summation contradict (20). All kernel denominators are positive because $\sigma_j > B \ge \sigma > 1/2$. □

Proposition 7.2 (Pointwise limitation of nonnegative weights)**.** *For fixed anchors, bounds, and nonnegative weights, the region (20) is contained in*

$$\bigcup_{j=1}^{m} \left\{ (\sigma, t) : \frac{1}{2} < \sigma \le B,\ \Phi_j(\sigma, t) > E_j \right\}.$$

Proof. If every $\Phi_j(\sigma, t) \le E_j$, their nonnegative weighted sum cannot exceed the weighted sum of the E_j. The contrapositive proves the assertion. □

Thus nonnegative weighting alone cannot create a pointwise exclusion beyond the union of the individual certified regions. Its role is to provide a common or compact certificate for a box or a finite cover. A real improvement beyond that union would require further information, such as unavoidable contributions from other zeros or constraints on a joint zero configuration. No such extra input is used here.

To extend the present geometry, one can first evaluate more exact anchors and take the actual union of their reflected-pair regions. Boxes inside this union can be certified by rational polynomial inequalities and finite covering arguments. This procedure gives local or finite-range statements; it does not by itself prove a prescribed improvement at every ordinate.

There is also a separate analytic challenge in extracting height-dependent improvements from the moment argument of [6]. Its internal truncation height is not the ordinate of a target zero, and a constant depending on a fixed target cannot automatically be converted into an improvement of an asymptotic exponent. A uniform curve such as $B - c/\log|t|$ would require new quantitative input relating the target ordinate to a strictly positive surplus in that argument. The present shape certificate makes no such inference.

Finally, a narrow high-ordinate rectangle can assist a specific contour deformation or a finite-range estimate. It does not lower a global error exponent in prime counting, since the boundary outside the certified windows is unchanged. The concrete contribution is the conditional quartic exclusion region, its exact closed rectangle, the reproducible external evaluation, and the finite real algebra verified in Lean.

References

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